{COLORIO}

CPU
PI0
PI1
COLORCS
MREAD1
IOCS
MRW
HFLIP
FIXCS
MREAD2
MLWR
MUWR
A[1..20]
D[0..15]
EXRES
DMAINIT
VFLIP
OCHAROMCS
OBJSCRH
OBJSCRV
OBJCS
TIM1
VBLANK
CPUCLK
CPURES
FIXDTAC
MINT16EN
DMACS
V1SCRH
V2SCRH
SVCHAROMCS
VRCSET
SCVRAMCS
SYSWR
MULWR
File: cpu.kicad_sch

Timing
DMACS
FIXCS
HFLIP
VFLIP
RST
P4H
{HVCOUNTERS}
X1S
VCNTEN
D0
CLK2
CLK21
CLK22
CLR
P1H
P2H
N2H
N2HP4H
8H
P256VD
TIM
TIM3
OBJBUFINIT
DMAINIT
DMACLR
OBJBUFCLR
OSCANCLR
FBREF
CSYNC
CBLK
CS1
CS2
RS1
RS2
OE1
OE2
HSYNC
WR
EXRES
VBLANK
CPUCLK
CPURES
FIXDTAC
1VD
SYNCRLDV
TIM1
RAS
DMA
File: timing.kicad_sch

Sprites1
File: sprites1.kicad_sch
CLK21
CLR
P1H
P2H
N2H
N2HP4H
8H
P256VD
TIM
TIM3
OBJBUFINIT
DMAINIT
DMACLR
OBJBUFCLR
OSCANCLR
FBREF
HFLIP
A[1..19]
D[0..15]
MLWR
MUWR
MREAD2
OCHAROMCS
OBJSCRH
OBJSCRV
OBJCS
DMAON
CRTKILL
DMAD

GFX – Fixed layer
CLK2
P1H
P2H
P4H
{HVCOUNTERS}
TIM3
HFLIP
FIXCS
FX[0..7]
MREAD2
MLWR
MUWR
A[1..11]
D[0..15]
File: GFX–Fixed layer.kicad_sch

Sprites 2
CLK21
CLK22
CLR
P2H
N2H
P256VD
X1S
VCNTEN
OCHF
FBWM1
FBWM2
VD[0..7]
HD[2..8]
HJ[0..1]
W[0..15]
VPD8
C[0..3]
CX[0..3]
HFLIP
VFLIP
CS1
CS2
RS1
RS2
OE1
OE2
HSYNC
WR
File: sprites2.kicad_sch

GFX – Tiles
CLK2
EXRES
CLR
P2H
P4H
N2H
P1H
N2HP4H
1V
VCNTEN
SYNCRLDV
TIM1
RAS
HFLIP
HSYNC
V1SCRH
V2SCRH
SVCHAROMCS
VRCSET
SCVRAMCS
SYSWR
MLWR
MUWR
MULWR
MREAD2
A[1..17]
D[0..15]
CRTKILL
1C[0..6]
2C[0..6]
File: GFX–Tiles.kicad_sch

Color mixer
CLK2
A[1..12]
D[0..7]
MLWR
FX[0..7]
OCD[0..7]
1C[0..6]
2C[0..6]
PI0
PI1
COLORCS
MREAD1
CBLK
CRTKILL
REDD
GREEND
BLUED
File: color_mixer.kicad_sch

IO
SPEAKER+
SPEAKER–
BSY
BUSY
RED
GREEN
BLUE
A[1..16]
D[0..7]
IOCS
EXRES
MRW
CSYNC
RST
MINT16END
DMAOND
CRTKILLD
SOUND–ON
SOUND–DATAD
SD0D
File: io.kicad_sch

Sound
EXRES
SOUND–ON
SOUND–DATA
SD0
D[0..7]
SPEAKER+D
SPEAKER–D
BSYD
BUSYD
File: sound.kicad_sch

Timing diagrams
File: timing_diagrams.kicad_sch

Sprite timing diagrams
File: sprite_timing_diagrams.kicad_sch

U216D 74LS04 M11
U228B 74LS21 E19
U205E 74LS07 H29
U197C 74LS07 A22
U205D 74LS07 H29
U205F 74LS07 H29
U226C 74LS08 J27
U226B 74LS08 J27
A23F 74LS07
U198A 74LS32 F15
U225B 74LS08 D30
U225A 74LS08 D30
U210A 74LS74 J23

Ulf Skutnabba, twitter: @skutis77
Konami GX870
Sheet: /
File: fround.kicad_sch
Title: The Final Round
Size: A3 | Date: 2024–09–08 | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 1/12

RA2
4.7k x 4
G31
MHALT
MRESET
VCC

RA5
4.7k x 8
E31
MLDS
MUDS
MAS
MRW
MDTACK
MBGACK
MBR
MBERR
VCC

U229
74LS367
H30
CPUCLK
GND
3-StateHexBuffers

VCC
G31
R48
1k

U222A
74LS74
F20
MINT16END
VBLANKD
GND
VCC

U122
68000D
E30
CLK
IPL0
IPL1
IPL2
MBGACK
BGACK
BG
BR
FC0
FC1
FC2
VMA
E
VPA
BERR
DTACK
HALT
RESET
AS
UDS
LDS
R/W
A1..A23
D0..D15

U224B
74LS00
G21
H28A
74LS04
MA23
MAS

U205A
74LS07
H29
U205C
74LS07
U205B
CPURESD

U228A
74LS21
E19
VOROMDTAC
FIXDTAC
J27
U226D
74LS08

U191A
74LS32
G24
XCS
MAS
A20
U191D
74LS32
G24

U227B
74LS74
E20
DMAINIT
OBJCS
U220C
74LS02
J21

U222B
74LS74
F20
SCVRAMCS
U204B
74LS02
H27
TIM1
U206E
74LS04
F19

MA[1..14]
A[1..20]
MD[0..15]

U88
27C010
F27
MA1..MA14
MD8..MD15
PGM
VPP
CE
OE
ROMCS

U139
G25
MUWR
R/W
CS2
MA13..MA9
MD15..MD8
6264SL
CS1
WORKCS

U137
27C010
E27
MD0..MD7

U105
E25
MUWR
6264SL
WORKCS

A[1..20]
D[0..15]

74LS367
U143
E26
3-StateHexBuffers

74LS245
U107
F24
buffers
A->B

74LS245
U96
F23
buffers
MREAD1

74LS245
U141
E24
buffers
MREAD1

U224A
74LS00
G21
ROMCS
WORKCS

MUWR
MULWR
MLWR
MRW
MREAD1
MREAD2
DMACS
SYSWR
VFLIP
HFLIP
PI0
PI1
EXRES

FIXCS 0x10 0000.. 0x1F FFFF Fixed layer RAM
SCVRAMCS 0x12 0000.. 0x13 FFFF Tile map 1 and 2 RAM
OBJCS 0x14 0000.. 0x15 FFFF Sprite RAM

U223B
74LS139
H23
A18
A17
SCVRAMCS
OBJCS
FIXCS
SCVRAMCS
OBJCS
decoder

H28F
74LS04
A20
U189A
74LS32
H24

U128
74LS138
address decoder
G23
A17
A18
A19
ROMCS
WORKCS
COLORCS
IOCS
VRCSET
SYSWR

U226A
74LS08
J27

U223A
74LS139
H23
MA22
MA21
MA23
XCS
SVCHAROMCS
OCHAROMCS
decoder

U191B
74LS32
G24
MAS

H28B
74LS04

U211C
74LS08
L25
CPUCLK

U209A
H21
74LS74

U209B
H21
74LS74
VOROMDTAC

U192B
74LS32
F21
MUWR
MULWR
MLWR

U192A
74LS32
F21

U211A
74LS08
L25

F19
U206B
74LS04
MREAD1
MREAD2

U206C
74LS04
F19

U188A
74LS32
K25
DMACS

ROMCS 0x0 0000.. 0x3 FFFF ROM
WORKCS 0x4 0000.. 0x5 FFFF Communication RAM
COLORCS 0x8 0000.. 0x9 FFFF Palette
IOCS 0xA 0000.. 0xB FFFF System I/O
SYSWR 0xC 0000.. 0xD FFFF System registers
(It can also be used for read) DMACS is activated)
VRCSET 0xE 0000.. 0xF FFFF Graphics bank

XCS 0x00 0000..0x1F FFFF
SVCHAROMCS 0x40 0000..0x5F FFFF Tile ROM
OCHAROMCS 0x60 0000..0x7F FFFF GFX ROM

U3
74LS138
K26
SYSWR
MULWR
OBJSCRH
OBJSCRV
V1SCRH
V2SCRH

U89
74LS273
O19
flip flops

CPU
Konami GX870
Sheet: /CPU/
File: cpu.kicad_sch
Title: The Final Round
Size: A3 | Date: 2024-09-08 | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 2/12

the four bits of each layer has the following meaning:
0000: Transparent
1111: Shadow
Else: Opaque

U180
63S141N
E16

U61
74LS153
Mux 4–1

U60
74LS153
Mux 4–1

U62
74LS153
Mux 4–1

U64
74LS153
Mux 4–1

U156
74LS174
hex flip flop

U121
74LS273
flip flops

U123
74LS157
S Quad Mux

U125
74LS157
S Quad Mux

U76
74LS157
S Quad Mux

U95
2128SL
HM6116P-3
2kB RAM

U117
2128SL
HM6116P-3
2kB RAM

U115
74LS273
flip flops

U94
74LS273
flip flops

U119
74LS245
buffers

U77
74LS245
buffers

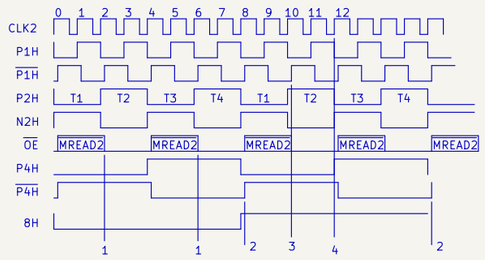
S1
A23
052535

S2
A20
052535

S3
A20
052535

Color Mixer	
Konami GX870	
Sheet: /Color mixer/	
File: color_mixer.kicad_sch	
Title: The Final Round	
Size: A3	Date: 2024-09-08
KiCad E.D.A. 10.0.4	Id: 3/12

CLK2: 6.1397MHz
1V: 8.00kHz
2V: 4.00kHz
4V: 2.000kHz
P1H: 3.0719MHz
P2H: 1.536MHz
P4H: 0.767971MHz
8H: 384.000kHz
16H: 192.000kHz



RAM Control

Slot Control
T1,T3 CPU
T2,T4 Video circuit

Time
1 Data output from RAM to Video circuit
2 Data latched into first DFF's
3 Data latched into second DFF's.
4 Pixel data output to FX[0..7]

The delay from the LSB (8H) of the RAM accessed by the video circuit is six pixels to output.

8-bits of data is latched every P4H cycle.
It is 2 pixels = 8bits / (4 bits / pixel).
The current pixel is multiplexed with P1H (also HFLIP).

B0 goes to pin 36 on timing circuit

Kolla 8V och 256H

When P2H is high I1 is selected.

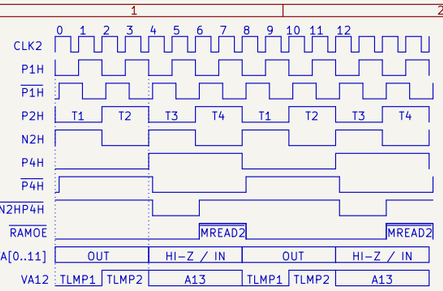
FXD[0..8]: Tile number
FXD[9..12]: Palette color
FXD[13]: HFLIP, not used on Final round
FXD[14]: VFLIP, not used on Final round

RAM
A[0..5] Horizontal position 0..511
A[6..10] Vertical position 0..255 in steps of 8
A[11.12] Tilemaps? Up to 4.

Fixed Layer
Konami GX870
Sheet: /GFX – Fixed layer/
File: GFX–Fixed layer.kicad_sch

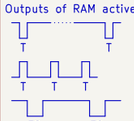
Title: The Final Round

Size: A3 | Date: 2024–09–08 | **Rev:**
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When VA is switched to output the bus is updated for each P1H cycle.

Outputs of RAM active when P4H is low, i.e during T1 and T2.



SYNCRLDV: Usually 60.604Hz
T is ~150ns.

VCNTEN: 8.0000kHz

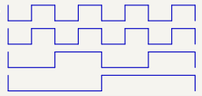
HSYNC: 16.0000kHz, T2: 5.2175us

Tilemap 1 and 2
VA[5..0]: Horizontal position: 0..63 -> 0..63 x 8 = 0..504
VA[11..6]: Vertical position: 0..63 -> 0..63 x 8 = 0..504
VA12 = 0: Tilemap 1, VA12 = 1: Tilemap 2

It is unclear what SYNCRLDV (Synchronized Reload Vertical?) and VCNTEN (Vertical Counter Enable?) are used for. They become active before HSYNC goes low. They might be used to synronize the output of the vertical counters so they are updated after a write to scroll registers before HSYNC goes low, and the vertical counters.

Data latched from the RAM:
Edge Tilemap
T1-T21
T2-T32

It is output 2 pixel clocks later to the ROMs.



The vertical counters are not the same value when they are offset with the scroll registers.

Address Register
0x06 V1SCRH Tilemap 1, horizontal scroll
0x08 V1SCRV Tilemap 1, vertical scroll
0x10 V2SCRH Tilemap 2, horizontal scroll
0x12 V2SCRV Tilemap 2, vertical scroll

V1SCRV and V2SCRV are in 007779 and are accessed when SYSWR and MULWR are active.

A total of 9-bits are used for vertical scroll.
2^9 = 512 scroll positions.

A13 = Hi: VRAM2(Tilemap 2).
A13 = Low: VRAM1(Tilemap 1)

V[13..15]: Color
V[11,12]: Tile bank
V[0..10]: Tile number

The 007781 is a line buffer. All counters related to all timing signals are inside the chip. It is synchronized on the rising edge of CLR and CLK2 after that.

1C[0:3]: Layer 1 Pixel color code
1C[4:6]: Layer 1 Palette
2C[0:3]: Layer 2 Pixel color code
2C[4:6]: Layer 2 Palette

Tile Layers
Konami GX870
Sheet: /GFX - Tiles/
File: GFX-Tiles.kicad_sch

Title: The Final Round

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 5/12

A[1..19]
D[0..15]

According to mame the object table for each frame buffer seems to be size 1800h = 6144 words. That is after processing 6144 / 80 = 76 sprites

OA[0..12]

K5
U43
007783

6264SL
U69 M10
6264SL
U71 M5

L14
U75
007785

O[0..17]

U23 74LS157 L21
U18 74LS157 M19
U30 74LS157 O6
U59 74LS157 O8
U32 O7

VPD8
VD[0..7]
HD[2..8]
HJ[0..1]
CX[0..3]
C[0..3]
FBWM2
FBWM1
OCHF
OSCNTEN

U8
K9
007784

OD[0..15]

74LS245
K20 buffers

74LS245
K21 buffers

BEFLAG
DMA
DMAINIT
OBJBUFINIT
MREAD2D
{ULRW}
{DMARW}

U189B 74LS32
U190C J25 74LS32

Object Color Data

U120 MX23C4100
U118 MX23C4100
AM-BITS(2M x 8/256K x 16) Depending on BYTE
D15/A-1
CMOS MASK ROM

SPA[0..17]

E[0..31]
D[0..15]

74LS86 U193D M10
74LS04 U216E M11
P2HD
OCHF

74LS257
U100 P10
U104 P11
U102 Q11
U98 Q10

OCHAROMCS

U55 74LS374 M9
U28 74LS374 M6
U15 74LS374 M8
U53 74LS374 M7

W[0..15]
N2HD

U210B 74LS74 J23

K10
U181 PAL16L8'
007793

RA9 2.2k x 8 L11
RA8 2.2k x 8 L6

007783
When BEFLAG goes active it will use the low byte OD[7..0] to create the address for the writes to sprite RAM.
According to mame:
OA[12..0] = 0x1800 | ((OD[7..0] & 0xff) << 2)

007784
– subtracts the scroll value for the sprites from current coordinates for the sprites.
– When DMA goes high. The 007784 will read 8 words, if bit OD15 of the first word is high then BEFLAG goes high and stays high until the sprite RAM has processed the first four words and possibly updated the four following words at the high end of the RAM.
it will compress the eigth words into four words and also subtract the x– and y–scroll values from the x and y coordinates. It will write the four words at the high end of the sprite RAM.
– Addresses at the low end goes from 0x0000..0x0007 and the following cycle at 0x0028..0x002F..

Scroll Register
OBJSCRV Sprite horizontal scroll
OBJSCRH Sprite vertical scroll

Data is latched in the scroll registers when the signals OBJSCRV and OBJSCRH go low.

The sprite memory is controlled in three separate cases:
1. OBJBUFINIT
2. BEFLAG
3. ~DMAINIT, when OBJBUFINIT and DMA are low.

007793

CLK21′ = CLK21

$\overline{LRW}$ = ~($\overline{DMAONS}$ & OBJBUFINIT & $\overline{CLK21'}$ +
BEFLAG & DMA & P1H & DMAONS & CLK21′ & (~8H & HFLIP + 8H & ~HFLIP)
~LWR & ~DMAINIT)

$\overline{URW}$ = ~($\overline{DMAONS}$ & OBJBUFINIT & $\overline{CLK21'}$ +
BEFLAG & DMA & P1H & DMAONS & CLK21′ & (8H & ~HFLIP + ~8H & HFLIP) +
~UWR & ~DMAINIT)

BEFLAG = Bus Enable FLAG?

OCHF: horizontal flip

007785
HJ[0..1] Unknown

C[0..3] Palette value for frame buffer 1
CX[0..3] Palette value for frame buffer 2

O[17:19]: RAM and ROM bank selects. Not used on final round.

If VPD8 is high, the vertical position is outside of visible area. The frame buffer is not updated then.

Sprite Layer – Part 1
Konami GX870
Sheet: /Sprites1/
File: sprites1.kicad_sch

Title: The Final Round

Size: A3	Date: 2024–09–08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/12

Pixel palette data

Frame Buffer 1

Frame Buffer 2

Pixel color data

Framebuffer 1 is outputting pixel data when P256VD is low and writes to the FB is possible when P256VD is high. The conditions are reveresed for FB2.

FBWM1: usually f= 250Hz, sometimes 1.15kHz
FBWM2: Mostly active all the time. When it is not active f = 60.06Hz with small active pulses, not synced to FBWM1. It becomes non active 25 horizontal lines after VBLANK goes active.

Writes to FB are possible when the "pixel part" is sleected from the mux.
If this expression is true (R[0:3] != 4'h0) && (FBWM1 | FBWM2 | ~WR | ~VPD8) data is written to DRAM.

Data is written when:
– WR is active.
– Not a transparent pixel (4'h0).
– FBWM1 and FBWM2 are not active: No writes to frame buffer going on.
– VPD8 (vertical counter highest bit, VCNT > 256) is not active. Outside visible area.

007787

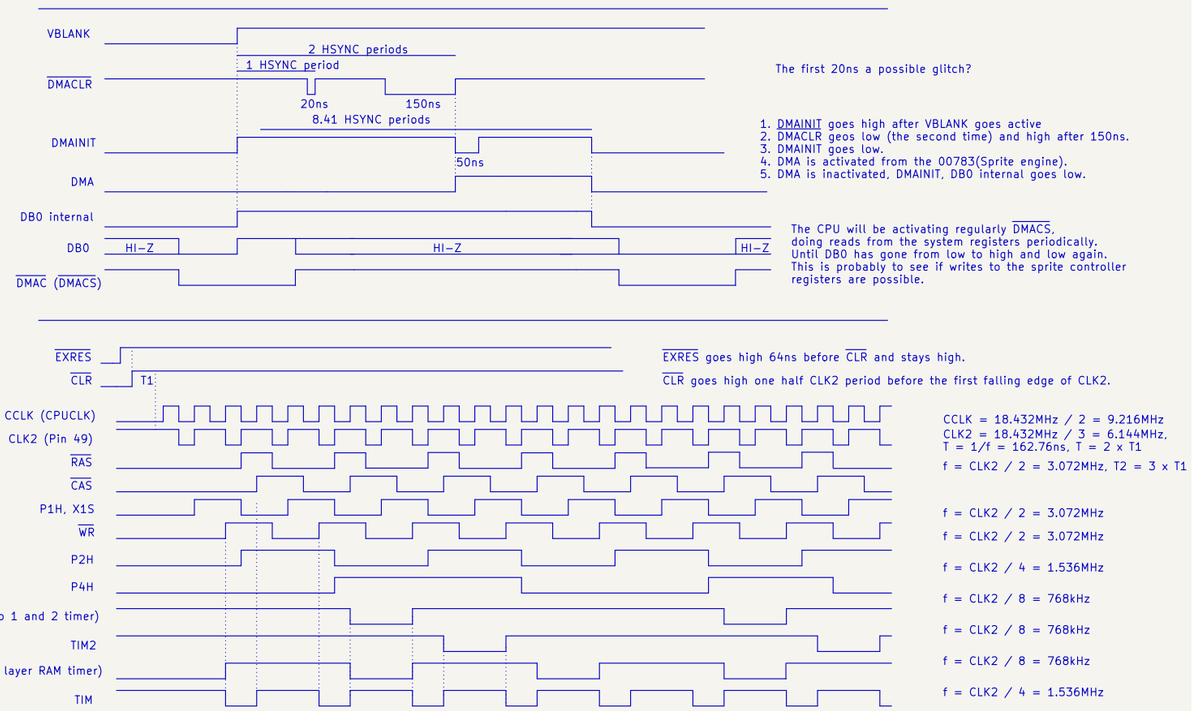
Multiplexer and buffer.
– When P256VD is low R[0:27] are the active inputs. When P256VD is high r[0:27]X are the active inputs.
– It will latch 4 nibbles (R[0:3], R[8:11]..) at a time and four CLK2 cycles later output one nibble at a time at the pixel clock rate.

Sprite Layer – Part 2
Konami GX870
Sheet: /Sprites 2/
File: sprites2.kicad_sch

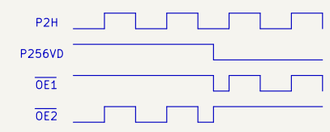
Title: The Final Round

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 7/12

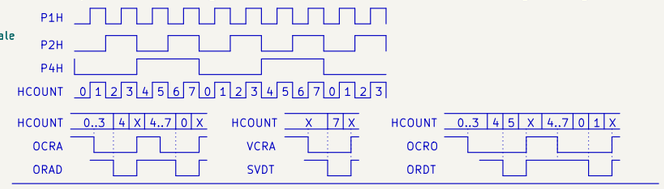
DMA Synchronization

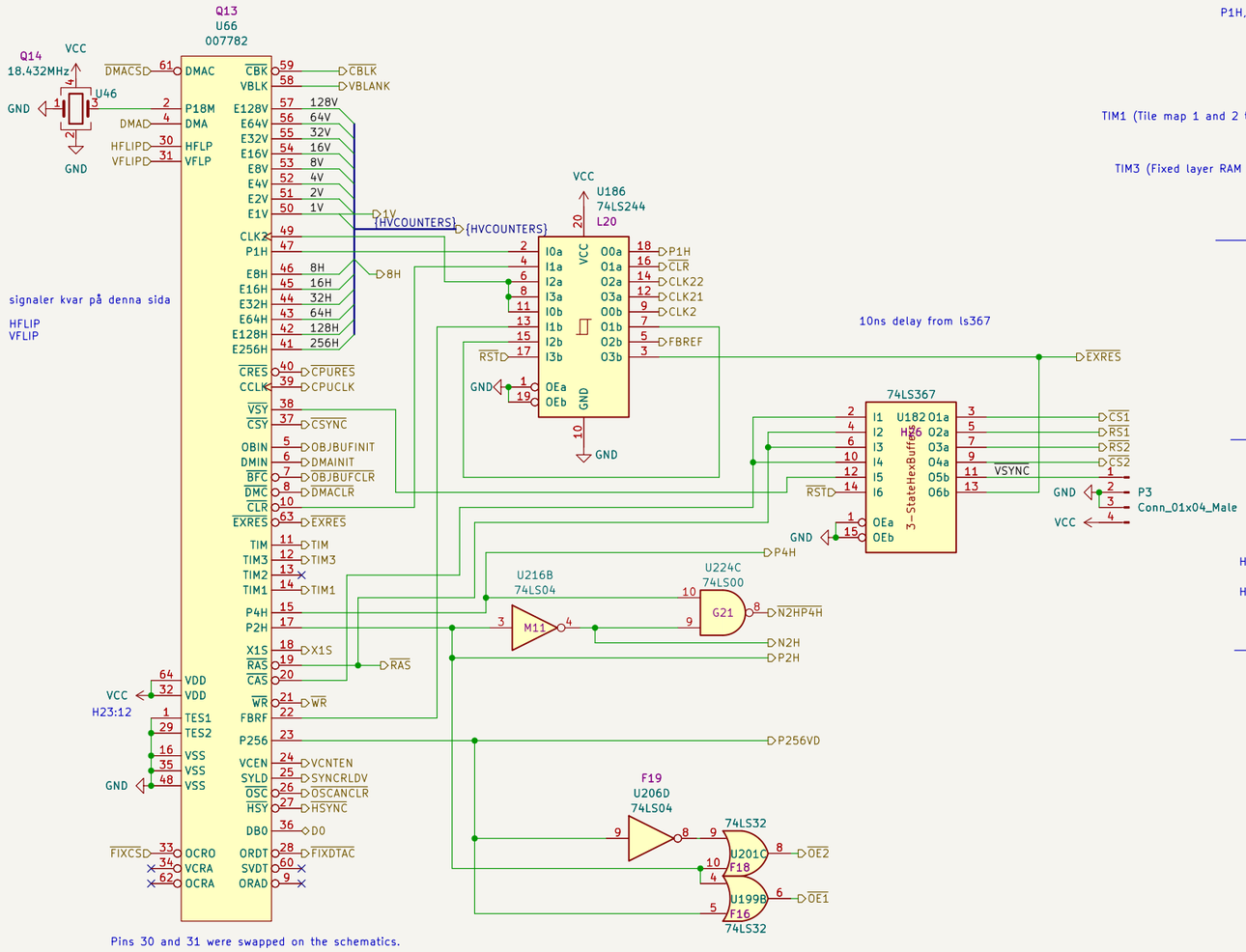


Output enables active vs. part of frame buffer active



Chip select signals and Data Acknowledge signals





Timing

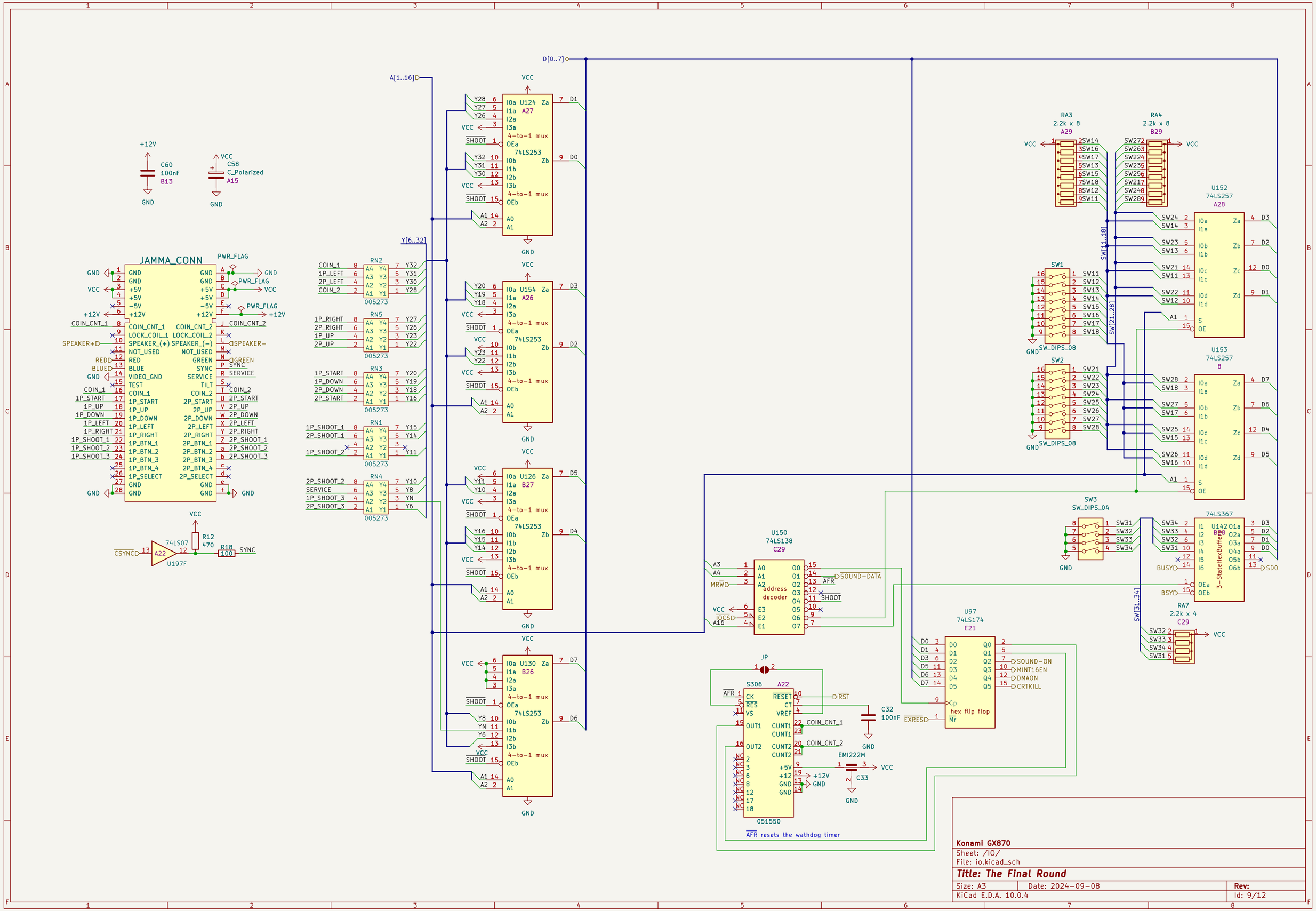
Konami GX870

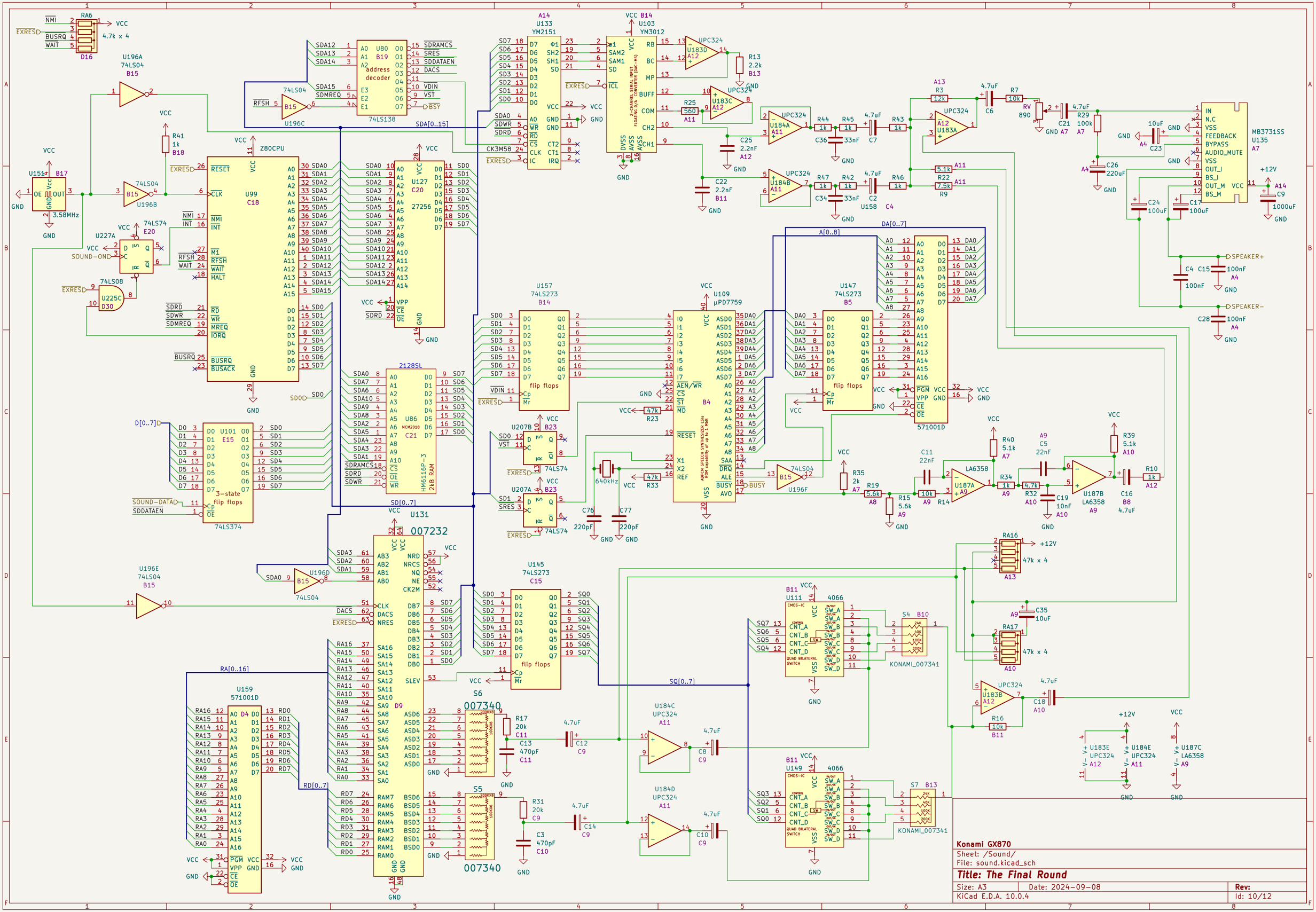
Sheet: /Timing/

File: timing.kicad_sch

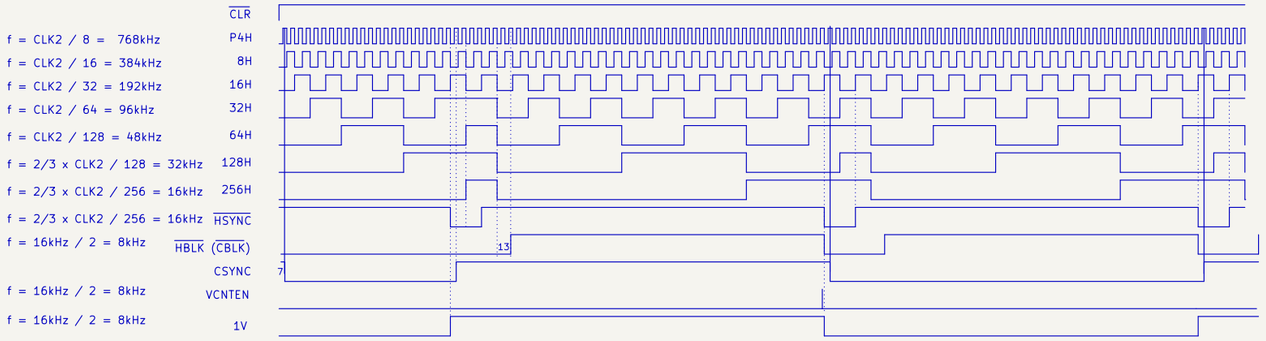
Title: The Final Round

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 8/12


D[0..7]
A[1..16]
Y[6..32]
+12V
C60
100nF
B13
GND
VCC
C58
C_Polarized
A15
GND
JAMMA_CONN
PWR_FLAG
GND
VCC
+5V
-5V
+12V
COIN_CNT_1
COIN_CNT_2
LOCK_COIL_1
LOCK_COIL_2
SPEAKER_(+)
SPEAKER_(-)
NOT_USED
RED
GREEN
BLUE
SYNC
VIDEO_GND
SERVICE
TEST
TILT
COIN_1
COIN_2
1P_START
2P_START
1P_UP
2P_UP
1P_DOWN
2P_DOWN
1P_LEFT
2P_LEFT
1P_RIGHT
2P_RIGHT
1P_BTN_1
2P_BTN_1
1P_BTN_2
2P_BTN_2
1P_BTN_3
2P_BTN_3
1P_BTN_4
2P_BTN_4
1P_SELECT
2P_SELECT
1P_SHOOT_1
2P_SHOOT_1
1P_SHOOT_2
2P_SHOOT_2
1P_SHOOT_3
2P_SHOOT_3
SPEAKER+
SPEAKER-
CSYNC
SYNC
R12
470
R18
100
74LS07
A22
U197F
RN1
RN2
RN3
RN4
RN5
005273
SERVICE
YN
74LS253
4-to-1 mux
SHOOT
A0
A1
U124 Za
A27
U126 Za
B27
U154 Za
A26
U130 Za
B26
D0
D1
D2
D3
D4
D5
D6
D7
U150
74LS138
C29
address decoder
SOUND-DATA
AFR
SHOOT
IOCS
A16
MRW
U97
74LS174
E21
hex flip flop
SOUND-ON
MINT16EN
DMAON
CRTKILL
EXRES
JP
S306
A22
RST
COIN_CNT_1
COIN_CNT_2
EMI222M
C33
C32
100nF
AFR resets the wathdog timer
051550
RA3
2.2k x 8
A29
RA4
2.2k x 8
B29
SW1
SW2
SW3
SW_DIPS_08
SW_DIPS_04
SW[11..18]
SW[21..28]
SW[31..34]
U152
74LS257
A28
U153
8
74LS367
U142
B28
3-StateHexBuffer
BUSY
BSY
SDO
RA7
2.2k x 4
C29
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RA6
NMI
EXRES
BUSRQ
WAIT
4.7k x 4
D16
VCC
U196A
74LS04
B15
R41
1k
B18
U151
B17
OE
OUT
3.58MHz
GND
U196B
U227A
74LS74
E20
SOUND-ON
74LS08
D30
U225C
U196C
RFSH
SDA12
SDA13
SDA14
SDA15
SDMREQ
U80
address
decoder
B19
74LS138
SDRAMCS
SRES
SDDATAEN
DACS
VDIN
VST
BSY
SDA[0..15]
Z80CPU
U99
C18
RESET
CLK
NMI
INT
M1
RFSH
WAIT
HALT
RD
WR
MREQ
IORQ
BUSRQ
BUSACK
U127
C20
27256
VPP
CE
OE
SDRD
SDWR
SDMREQ
SD0
SD1
SD2
SD3
SD4
SD5
SD6
SD7
U133
YM2151
A14
SH1
SH2
SO
SI
IC
CK3M58
U103
YM3012
2-CHANNEL SERIAL INPUT FLOATING D/A CONVERTER (DAC-HS)
SAM2
SAM1
SD
ICL
RB
BC
MP
BUFF
COM
CH2
CH1
DVSS
AVSS
UPC324
U183D
U183C
U184A
U184B
R13
2.2k
B13
R25
560
A11
C25
2.2nF
A12
C22
2.2nF
B11
R44
R45
R47
R42
1k
33nF
C36
C34
C7
C2
U158
C4
R43
R46
R3
12k
R7
10k
R22
5.1k
R9
7.5k
C6
4.7uF
RV
890
R29
100k
A7
C21
C26
220uF
A4
C23
10uF
C24
100uF
C17
100uF
MB3731SS
U135
A7
IN
N.C
VSS
FEEDBACK
BYPASS
AUDIO_MUTE
OUT_I
BS_I
OUT_M
BS_M
+12V
C9
1000uF
SPEAKER+
SPEAKER-
C4
C15
C28
100nF
A4
DA[0..7]
A[0..8]
U147
74LS273
B5
flip flops
571001D
PGM
VPP
CE
OE
U157
74LS273
B14
U109
µPD7759
ASD0
ASD1
ASD2
ASD3
ASD4
ASD5
ASD6
ASD7
AEN/WR
CS
ST
MD
RESET
X1
X2
REF
ADPCM SPEECH SYNTHESIZER LSIs ROM capability of up to 1 Mbit
B4
640kHz
C76
220pF
C77
220pF
R23
47k
R33
47k
U207B
U207A
B23
VDIN
EXRES
SD[0..7]
U131
007232
NRD
NRCS
NQ
NE
CK2M
DB7
DB6
DB5
DB4
DB3
DB2
DB1
DB0
SLEV
DACS
NRES
SA16
SA15
SA14
SA13
SA12
SA11
SA10
SA9
SA8
SA7
SA6
SA5
SA4
SA3
SA2
SA1
SA0
D9
U145
74LS273
C15
SQ0
SQ1
SQ2
SQ3
SQ4
SQ5
SQ6
SQ7
SQ[0..7]
RA[0..16]
RD[0..7]
U159
571001D
RD0
RD1
RD2
RD3
RD4
RD5
RD6
RD7
S6
S5
007340
200KB
R17
20k
C11
470pF
C13
R31
20k
C9
C3
470pF
C10
C12
C14
4.7uF
U184C
U184D
UPC324
A11
C8
C9
C10
U111
U149
4066
CMOS-IC
QUAD BILATERAL SWITCH
CNT_A
CNT_B
CNT_C
CNT_D
SW_A
SW_B
SW_C
SW_D
S4
S7
B10
B13
KONAMI_007341
24K
40K
100K
200K
RA16
RA17
47k x 4
A13
A10
C35
10uF
A9
R16
10k
B11
C18
A10
U183B
U183E
U184E
U187C
LA6358
A9
A12
R35
2k
A7
R19
5.6k
A8
R15
5.6k
A9
C11
22nF
R14
10k
R40
5.1k
R34
1k
C5
22nF
R32
4.7k
C19
10nF
A10
R39
5.1k
A10
R10
1k
A12
C16
B8
4.7uF
R12
12k
A13
R24
5.1k
R26
7.5k
R27
10k
C29
100nF
C30
33nF
C31
4.7uF
C32
220uF
C33
10uF
C34
100nF
R28
1k
R30
1k
C14
100nF
C27
33nF
C28
4.7uF
C29
220uF
C30
10uF
C31
100nF
R29
1k
R31
1k
C32
33nF
C33
4.7uF
C34
220uF
C35
10uF
C36
100nF
R36
1k
R37
1k
R38
1k
R39
1k
R40
1k
R41
1k
R42
1k
R43
1k
R44
1k
R45
1k
R46
1k
R47
1k
R48
1k
R49
1k
R50
1k
R51
1k
R52
1k
R53
1k
R54
1k
R55
1k

Horizontal signals

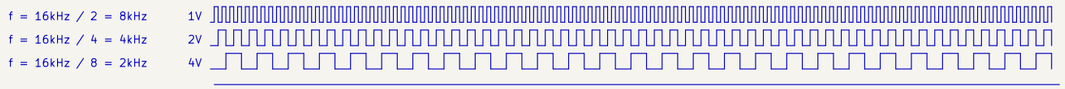


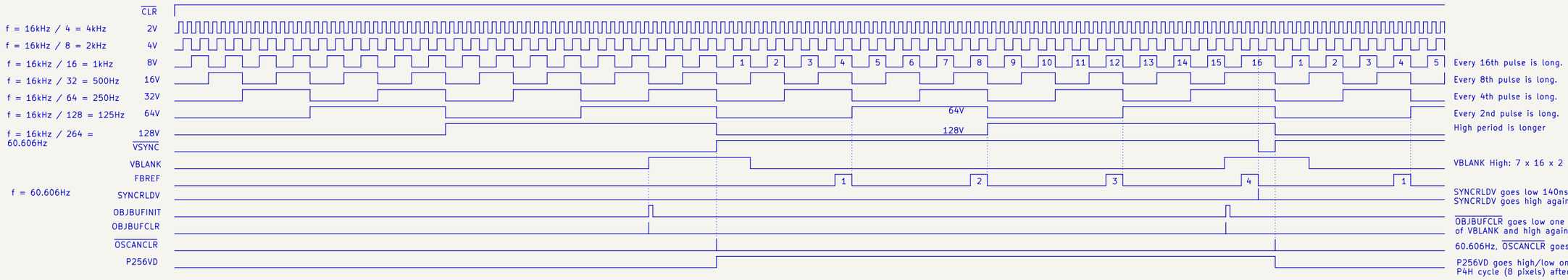
– The first VCNTEN is skipped after reset.
It goes low 140ns before HSYNC goes low,
and high again when HSYNC goes low.
VCNTEN is active right before every second falling edge of HSYNC.

– CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

The first HSYNC is at HCOUNT 175 after reset.

Vertical signals





FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

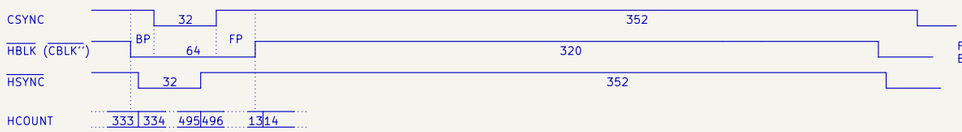
Horizontal and vertical synch timing diagrams

Pixel clock CLK2 = 18.432MHz / 3 = 6.144MHz

Horizontal lines
HSYNC freq = 6.144MHz / 384 = 16kHz
HSYNC period = 1 / 16kHz = 62.5us
Pixels / line: 384
Active pixels: 320
Blank pixels: 64

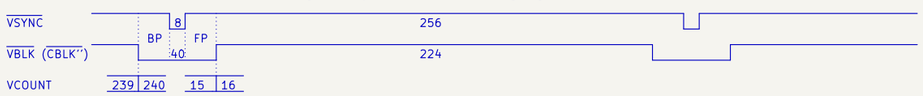
Vertical lines
VSYNC freq = 16kHz / 264 = 60.606060Hz
VSYNC period = 1 / f = 264 / 16kHz = 16.5ms
scanlines / frame: 264
Active lines: 224
Blank lines: 40

The numbers in the HSYNC and HBLK diagrams are HSYNC cycles.
All edges are synchronised to the rising edge of CLK2.



Front Porch (FP) 24 px,
Back Porch (BP) 8 px

The numbers in the VSYNC and VBLK diagrams are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



Front Porch (FP) and
Back Porch (BP) 16 HSYNC cycles.

HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

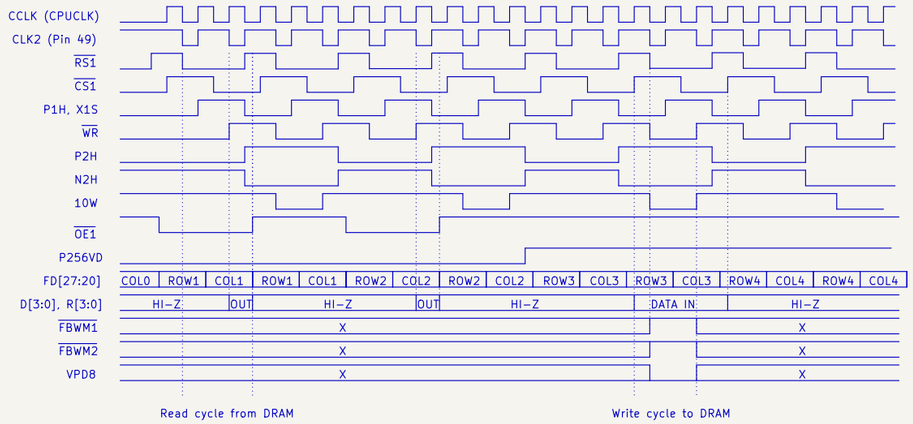
CBLK" is at the output of color mixer.

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The same row and column addresses are repeated twice.
One for read cycle and the other for write cycles.

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